

Reclaiming Play in Digital Mental Health: A Scoping Review of Play-Oriented Gamification for Stress in College Students

Rowan Stratton¹[0009-0005-7976-869X] and Jomara Sandbulte¹[0000-0002-7798-2853]

University of Minnesota Duluth, Duluth MN , USA
strat238@d.umn.edu
jsandbul@d.umn.edu

Digital mental health (DMH) applications frequently incorporate gamification to encourage engagement with digital wellness support, often through visual trackers or reward-based aesthetics. However, many of these systems rely on structured incentive mechanisms that may not align with how play is meaningfully experienced. This scoping review examines how play and gamification are framed within DMH design and what limitations are reported in the literature. Following PRISMA-ScR guidelines, we reviewed 66 empirical studies published between 2013 and 2025 and qualitatively synthesized how engagement and perceived benefits were described. Across the reviewed studies, points, streaks, and progression mechanics dominated design approaches and were frequently discussed in relation to reported pressure, disengagement, or diminished meaning over time. Fewer studies described play-oriented approaches emphasizing narrative exploration, self-expression, or intrinsic motivation. Even when imaginative or exploratory framing was present, these elements were often diluted by engagement tactics that shifted interaction toward compliance rather than play. We contribute a play-based reframing of DMH gamification and highlight design opportunities for systems that support agency, interpretation, and meaning-making beyond incentive-driven engagement.

Keywords: Digital mental health, gamification, play theory, playful interaction, scoping review, interaction design, meaning-making

1 Introduction

Across college campuses in the United States, stress can be part of many people's everyday experience. This condition can be pervasive and consequential to surrounding factors, and for college students, these factors are susceptible to shifts [44]. National surveys consistently identify stress as one of the most common factors negatively impacting academic performance among students [5, 35]. For those impacted, elevated stress has been associated with increased dropout risk, academic impairment, and poorer mental health outcomes [35, 68]. Reported downstream effects of this condition without proper care can impact sleep, nutrition, social engagement, and overall well-being [5].

Although evidence-based therapies can be effective in addressing stress-related symptoms, access to sustained mental health care remains limited for many college students [19]. Fewer than 40% of students who report a need for mental health support receive ongoing care [35]. Structural barriers such as scheduling conflicts, academic workload, cost, and limited service availability continue to constrain access, motivating interest in alternative or supplemental forms of support within higher education [19, 34, 46].

Digital mental health (DMH) applications have emerged as a prominent approach for delivering stress-management and psychological support tools [33]. These systems can translate therapeutic practices, including mindfulness, mood tracking, and cognitive behavioral techniques, into digital formats that offer flexibility, privacy, and asynchronous access [8]. While some DMH interventions demonstrate outcomes comparable to in-person treatment, these benefits are most consistent when users adhere to prescribed activities [6]. Sustaining engagement, however, remains a persistent challenge, with high attrition and inconsistent use limiting long-term effectiveness [45, 46].

To address engagement challenges, gamification has become a widely used strategy within DMH applications [14, 18, 28]. Gamification introduces selected game elements—such as points, achievements, or streaks—into non-game contexts to motivate continued use [18, 56]. Evidence for these approaches is mixed [17, 55]. Prior work suggests that reward-driven mechanics may be initially appealing, emphasizing compliance through achievement or goal tracking, enabling some to visualize progression [41, 42]. In the context of mental health support, this may introduce pressure or self-evaluative stress [24, 30, 59].

Due to the reported effects of games on stress and mental health across disciplines [17, 21], existing reviews demonstrate sustained interest in game-informed mental health interventions. This interest evaluates the aspects of games that leverage our sense of play and motivations for tasks of self-care into forms such as serious games and gamified DMH tools [14, 28, 64]. However, much of this literature prioritizes adherence or short-term engagement, with comparatively less attention to play as an experiential process supporting meaning-making, reflection, or emotional regulation [43, 58, 61]. While gamification research often evaluates success through usage metrics, play theory is a more transient state that emphasizes experiential qualities such as interpretation, agency, and emotional resonance [43, 58, 60]. The distinction between gamification mechanics and playfulness as lived experience remains underdeveloped within digital mental health design [33, 43].

To address this gap, we conducted a scoping review to systematically map how game elements and play-oriented structures are employed and discussed in smartphone-based digital mental health applications for college students [7, 31, 51, 63]. Our review examines dominant design patterns, ethical tensions, and underexplored opportunities for supporting emotionally meaningful and sustainable stress-management practices.

The following research questions guide this review:

1. RQ1: Which game elements grounded in play theory are most commonly used in mental health applications for college students?
2. RQ2: What ethical and design challenges arise when integrating these elements into stress-management interventions?
3. RQ3: How do gamified mental health applications frame playfulness in ways that may influence stress management and reported benefits among undergraduate students?

By synthesizing existing literature, this review clarifies the strengths and limitations of current gamified mental health applications and outlines design considerations for HCI researchers and practitioners seeking to support ethical, reflective, and emotionally sustainable engagement in digital stress-management tools.

2 Methodology

This scoping review investigates existing literature on the use of gamified digital mental health (DMH) applications designed to support stress management in higher education contexts. The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines [63]. The review protocol was developed a priori following established scoping review methodologies [7].

2.1 Search Strategy

The search strategy followed established scoping review methodologies [7, 31, 63]. Between September 2024 and February 2025, systematic keyword searches were conducted across eight electronic databases: ACM Digital Library, Scopus, IEEE Xplore, Taylor & Francis, EBSCO, ScienceDirect, and Springer.

Search strings combined terms related to gamification, digital mental health, stress management, play, and higher education. Boolean operators (AND, OR) and truncation were used to capture variations of key terms (e.g., `gamif*`). Searches were restricted to peer-reviewed publications written in English and published between January 2010 and November 2024. This timeframe was selected to reflect contemporary developments in mobile computing, digital mental health, and interactive system design.

2.2 Search Terms

Search queries were constructed to reflect three primary conceptual domains: the target population (college or undergraduate students), the technological approach (gamification, games, or play-informed digital systems), and the psychological focus (stress or stress-related mental health outcomes). Keywords were informed by prior reviews of digital mental health interventions, gamification in health contexts, and play-oriented design [8, 14, 28]. The complete set of database-specific search queries is provided in Appendix A2 (Table 4).

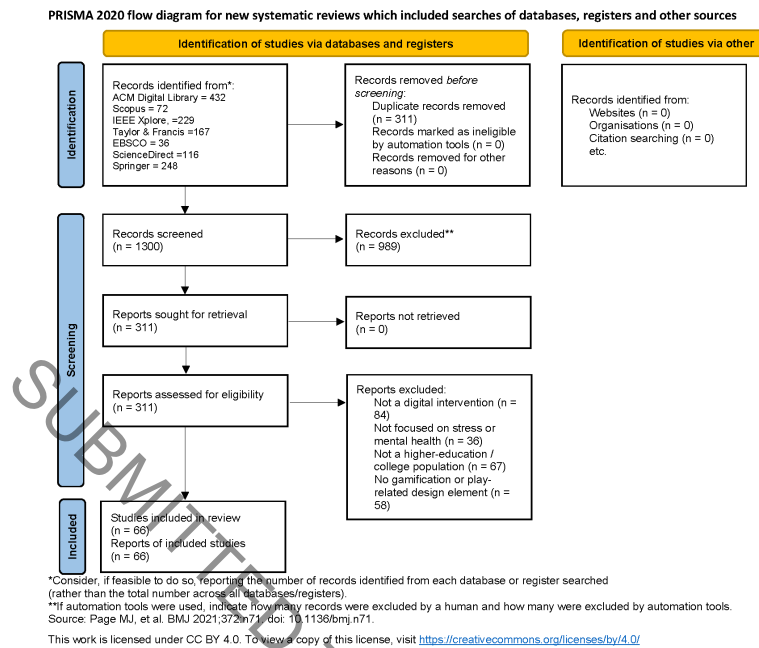


Fig. 1. PRISMA flow diagram illustrating the study selection process for the scoping review, including identification, screening, eligibility assessment, and final inclusion of studies.

2.3 Eligibility Criteria

Studies were eligible for inclusion if they met all of the following criteria. First, the study examined a digital mental health application designed to support stress management or mental well-being among college or university students. Second, the application incorporated gamified or play-oriented design elements, such as reward systems, progression mechanics, narrative structures, or playful interaction techniques. Third, the intervention was designed for or evaluated on smartphones or mobile devices.

Studies were excluded if they focused on entertainment games not intended for mental health support, serious games developed as standalone games rather than applications, or non-digital interventions. Studies targeting populations outside of higher education were also excluded. Only full research articles published in peer-reviewed journals or conference proceedings were considered. Reviews, meta-analyses, mapping studies, editorials, and position papers were excluded. Articles had to be written in English and available in full text. No restrictions were placed on study design, outcome measures, country of origin, or publication venue.

Table 1. Synthesized engagement framings identified across reviewed digital mental health interventions, grounded in the prevalence of reported game mechanics rather than mutually exclusive study classifications.

Engagement Framing	Common mechanics reported (number of studies)	Reported limitations
Traditional Gamification	Progress tracking (30), badges (26), points (25), streak tracking (15)	Novelty effects; disengagement as reward value diminishes
Personalized Gamification	Goal-setting (28), adaptive or user-tailored challenges (reported variably)	Still relies on structured engagement loops
Social Gamification	Leaderboards (15), collaborative features (5)	Pressure or stress for some users due to comparison
Play-Oriented Framing (identified in this review)	Narrative components (10), quest or mission structures (13), avatar customization (7)	Underexplored in digital mental health literature

2.4 Study Selection

Study selection was conducted in two phases. In the first phase, titles and abstracts were screened to assess relevance to the inclusion criteria. To ensure consistency in interpretation, articles assessed were discussed among both authors to mitigate discrepancies in screening decisions. Any identified discrepancies were discussed and resolved through consensus.

In the second phase, full-text articles were assessed for eligibility based on the predefined criteria. Reasons for exclusion at this stage were documented and are reported in the PRISMA flow diagram (see Figure 1).

2.5 Data Extraction and Synthesis

Data extraction was conducted by the first author and verified by the second author. Extracted information included study characteristics, target population, application features, gamification or play elements, theoretical framing, and reported outcomes related to stress or well-being. Extracted data were recorded in a shared spreadsheet and synthesized using a descriptive analytical approach. The data search table is provided in Appendix A to demonstrate this extraction process and Table 1 outlines how the work was synthesized.

3 Results

A total of 66 studies met the inclusion criteria illustrated in Fig 1. Initially, 1611 records were retrieved from the selected databases (see Section 2.1). After the

screening procedure (see Section 2.3), 311 records were retained for further consideration. A citation search was then performed, and titles and abstracts of cited articles were screened. Following full-text review, 66 studies were included in this review. Across these studies distinct gamified DMH applications were identified, targeting stress management (see Appendix C for alignment summary). The results are organized using Whitton’s framework of play (see Table 2) to examine how reported mechanics were framed across studies and how authors discussed potential benefits and limitations.

3.1 Mechanics Reported Across the Corpus

For this review, gamification was classified based on game mechanics commonly identified in prior literature, drawing on frameworks described by Deterding et al. [18]. When studies used different terminology to describe mechanics with similar functional roles, descriptions of interaction and system behavior were used to group mechanics into shared categories. For example, mechanics described as quests designed to reach a target outcome were grouped with goal-setting features when both served comparable functional purposes (see Table 1).

Across the included studies, the most frequently reported mechanics were progress tracking (30 studies), goal-setting (28), badges (26), points (25), levels (19), and leaderboards (15). Narrative components appeared in (10) studies, quests in (13), avatar customization in (7), chatbot interactions in (6), and collaborative features in (5). Table 2 summarizes the number of studies that reported each mechanic and the type of play each categorically aligns with [65]. Because many interventions incorporated multiple mechanics within a single application, individual studies often contributed to multiple categories.

3.2 Play-Based Grouping of Design Patterns

Using the play framework, each of the identified mechanics was grouped into key characteristics in which individuals could explore items such as failure, immersion, and intrinsic motivation [66]. These are defined as characteristics that enhance the experience of play [65]. Taking this identified framework in education that evaluates the philosophical game ideal on the role of the ‘*magic circle*’ in play, this work aims to work to embrace similar philosophies [43, 60]. To interpret how these mechanics were used and framed in alignment with the framework, we grouped reported features and interaction patterns in alignment with characteristics of the framework [65]. This framework distinguishes various forms of playful activities that can range from structured and rule-bound to exploratory and imaginative. Appendix C outlines the details of the summarized results of elements aligned with each characteristic of play.

3.3 Structured Oriented Play

Structured-oriented play was a consistent pattern across the corpus. 46 out of 66 studies (approximately 70%) implemented extrinsic progress-oriented mechanics

Table 2. Relative Prevalence of Game Elements Across Whitton’s Play Types (n=66)

Game Element	Structured	Exploration	Imaginative	Social
Points (25)	•••	—	—	—
Badges (26)	•••	—	—	—
Progress Tracking (30)	•••	—	—	—
Streak Tracking (15)	••	—	—	—
Daily Check-ins (12)	—	••	—	—
Chatbot Interaction (6)	—	•	—	—
Quest / Mission Structure (17)	••	••	•	—
Avatar Customization (7)	—	—	•	—
Cooperative Support Quests (9)	—	—	—	•
Competitive Social Features (15)	—	—	—	••

Note. Dot density represents prevalence bands across included studies (•= 1–10, ••= 11–20, •••= 21+). Elements may appear across multiple play types depending on framing. Dots indicate presence across categories rather than partitioned counts.

such as badges, points, streaks, progress indicators, or comparative feedback. Within this subset, 10 out of 66 studies (approximately 15%) explicitly reported short-term increases in participation following the introduction of visible rewards or social comparison features [3, 30, 41].

Beyond explicitly reported outcomes, 18 studies (approximately 27%) were coded as exhibiting patterns consistent with what prior work characterizes as a checklist effect, in which participation centers on maintaining visible progress or satisfying system prompts rather than supporting reflective or meaning-oriented interaction [27, 42, 57]. This framing was most commonly reported in mood tracking and CBT-oriented interventions [6, 26].

Specific mechanics associated with this pattern included **streak tracking**, which appeared in 15 studies (approximately 23%). These studies commonly described streaks as mechanisms intended to encourage short-term adherence [22, 55]. However, 12 out of these 15 studies also reported declining engagement over time, which authors frequently attributed to novelty effects or reduced motivation following initial adoption [40, 55]. Several studies further reported that participants experienced streak loss as introducing pressure or stress, particularly when participation was framed around maintaining performance [2, 23, 29, 42].

Progress tracking and goal-oriented incentives were also prevalent within structured designs. Progress indicators appeared in 30 studies (approximately 45%), goal-setting in 28 studies (approximately 42%), and rewards-based incentives in 29 studies (approximately 44%) [4, 15]. Authors frequently noted that these structures supported compliance with repeated actions, while simultane-

ously raising concerns that engagement could become superficial when interactions were reduced to repeated logging or completion of system prompts [32, 57].

3.4 Exploration-Oriented and Self-Directed Interaction

A smaller subset of studies emphasized exploration-oriented interaction, including narrative progression, self-directed quests, and open-ended reflective engagement [65, 66]. These designs gave users greater flexibility in how and when they engaged with the intervention. Narrative components appeared in *10* studies (approximately *15%*), quests in *13* studies (approximately *20%*), and collaborative features in *five* studies (approximately *8%*). When present, these elements were described by authors as framing stress management as an ongoing process rather than as a sequence of required tasks [13, 58, 66].

Daily check-ins appeared in *12* studies (approximately *18%*). In studies that framed check-ins as low-pressure reflection rather than performance tracking, authors framed these interactions as more self-directed [10, 52]. **Chatbot interactions** appeared in *six* studies (approximately *9%*) and were commonly described as blending structured guidance with conversational choice [12, 20]. In one narrative intervention, campus-themed quests and interactive storytelling were discussed as fostering intrinsic motivation compared to simpler tracking structures [13].

Because this is a scoping review, these observations are treated as reported patterns in design and outcomes rather than as evidence of causal effects. Across the corpus, exploration-oriented structures were less frequently reported than structured reward-loop designs. These insights aim to reflect trends identified as present in DMH applications [14, 55].

3.5 Imagination Play and Symbolic Representation

Imagination-oriented mechanics emphasizing symbolic representation and meaning-making appeared in *10* studies (approximately *15%*). These designs commonly used narrative missions, metaphor, or creative interaction which authors described as supporting the interpretation of stress-related experiences [10, 58, 65]. Within this subset, narrative-driven missions such as those used in *SPARX* were frequently discussed by authors in relation to reflection and continued engagement [48, 66].

Avatar customization appeared in *seven* studies (approximately *11%*) and was discussed as supporting personal expression through a digital self [12, 20]. Studies examining narrative missions, avatar customization, and other imagination-based mechanics discussed symbolic interaction in relation to autonomy and identity-relevant motivational processes [10, 58]. However, *three* studies reported that imagination-based mechanics could be perceived as time-consuming, with some participants preferring quicker interactions or immediate incentives [10, 32, 58]. Overall, imagination-oriented play was reported less frequently than structured reward-loop approaches.

3.6 Social Play and Cooperative Support

Social play mechanics varied across the corpus, particularly depending on whether they emphasized cooperation or competition. *Nine* studies (approximately 14%) implemented cooperative structures such as shared goals or social support quests. These studies discussed cooperative mechanics in relation to emotional support and accountability [1, 36]. Competitive social structures were more frequently discussed in relation to performance pressure or negative comparison [29, 36].

Across studies that discussed social play, cooperative features were more often framed as supportive, while competitive features were more often discussed as pressuring. Within the subset using social support quests, *2 out of 9* studies reported early participation increases that were sustained in relation to shared goals and supportive bonds. Cooperative designs were less common than competitive or comparative mechanics across the full corpus.

3.7 Ethical and Cultural Considerations

Ethical concerns related to privacy and autonomy were explicitly discussed in *eight* studies (approximately 12%) [50, 54]. Within this subset, *5 out of 8* studies reported concerns that publicly visible mechanics, such as leaderboards or progress sharing, could unintentionally expose emotional states or coping progress. Authors described this as contributing to discomfort or stigma in some contexts. These studies emphasized the importance of informed consent, user-controlled visibility settings, and clear data governance for mental health applications [30, 40, 69].

A separate concern was over-engagement, discussed in *seven* studies [37, 38, 59]. These papers described cases where competitive or highly immersive structures were associated with short-term motivational benefits but also raised concerns about compulsion, over-reliance, or participation that conflicted with offline responsibilities.

Cultural sensitivity was discussed in *six* studies (approximately 9%) [9, 17, 59]. *Six studies* reported misalignment between individual achievement mechanics and collective norms in some contexts, which was discussed in relation to reduced engagement or increased stress. Conversely, *2 out of 6* studies described improved fit when mechanics were culturally tailored, such as framing progress through shared values or group-based goals.

4 Discussion

This scoping review highlights a consistent pattern across DMH interventions for college students: many systems from the sample relied on extrinsic mechanics such as points, badges, streak counters, and progress tracking to structure engagement. Across the reviewed studies, these features were frequently discussed in relation to short-term increases in participation, but were less consistently framed as supporting sustained engagement or longer-term stress-management

practices [33, 38, 41]. In contrast, a smaller subset of studies described play-oriented approaches as enabling experiences that participants perceived as more personally meaningful, particularly when interaction supported reflection and autonomy [43, 58].

Across the corpus, the distinction between these approaches was not the presence of game mechanics alone, but how those mechanics framed participation and emotional engagement. Interventions that combined immediate feedback with reflective or narrative elements were more often discussed by authors as meaningful or supportive over time than those relying exclusively on reward loops [29, 43]. At the same time, several studies cautioned that narrative or experiential components were frequently underdeveloped or insufficiently tailored to the target audience, indicating that playfulness does not emerge simply from adding story elements [11, 33]. Together, these findings suggest that play-oriented design in digital mental health hinges less on specific mechanics and more on how interaction structures agency, interpretation, and emotional meaning-making [57, 58, 66].

4.1 Design Implications

Across the reviewed literature, reward-oriented approaches were frequently associated with short-term increases in activity, but were less consistently discussed as supporting deeper emotional change when used in isolation [42, 55]. In contrast, Whitton’s play framework emphasizes imaginary and symbolic play within a bounded playspace, where participants can explore meaning and emotion through creative interaction rather than performance signaling [65, 66]. This framing aligns with therapeutic traditions that emphasize reflection, reinterpretation, and rehearsal. Within games this could translate to narrative or decision-based approaches without requiring constant evaluation or visible progress [20, 43, 58].

From a design perspective, this suggests that play-oriented digital mental health systems should prioritize how interaction structures agency, interpretation, and emotional engagement, rather than focusing narrowly on sustaining activity. Several studies emphasized that experiential qualities of interaction—such as flexibility, narrative framing, and symbolic meaning—were central to how users made sense of engagement over time [43, 66]. These findings indicate that the impact of play-based design may be better understood through evaluative approaches that extend beyond behavioral metrics alone.

Accordingly, evaluation methods that foreground lived experience are particularly relevant for assessing play-oriented systems. Experience sampling can capture in-the-moment affect and context [12, 46], while diary studies support understanding engagement trajectories and perceived value beyond initial novelty [47, 67]. Qualitative approaches, including interviews and thematic analysis of reflective accounts, further illuminate how users interpret play and integrate it into coping practices over time [66]. Together, these methods help align evaluation strategies with the reflective outcomes emphasized by play theory.

4.2 Moving Toward Playful Design

Building on these implications, our review suggests that designers should move beyond an exclusive focus on sustaining activity through rewards and instead support interaction patterns that allow user-controlled pacing, optional narrative framing, and non-evaluative forms of engagement. Prior work indicates that reward-driven gamification often emphasizes compliance and short-term participation rather than emotionally meaningful interaction [14, 28, 42]. In contrast, play-oriented approaches emphasize autonomy, interpretation, and experiential engagement, which may better support flexible and personally resonant stress-management practices [27, 57, 58].

Design patterns such as role-based scenarios, narrative quests, and reflective challenges offer concrete ways for users to rehearse coping strategies through play, particularly when stressors are framed as situations to explore rather than tasks to complete [39, 49]. Rather than centering success on points, badges, or streak continuity, these approaches shift attention toward coping practices that remain meaningful when external rewards lose novelty [16, 42]. Whitton’s play framework provides a useful lens for distinguishing mechanics that primarily promote compliance from those that more strongly support agency, interpretation, and meaning-making [12, 66].

4.3 Ethical Considerations

The review highlights ethical challenges associated with designs that rely exclusively on extrinsic rewards and public performance indicators. Leaderboards and visible progress displays may impose pressure, exacerbate social comparison, and increase privacy risk, particularly in mental health contexts where participation can signal vulnerability [38, 42, 59]. At the same time, some users report valuing group-based features that support accountability or reduce isolation when participation and data visibility are carefully controlled [50, 53]. These divergent preferences reinforce the importance of providing adjustable interaction modes and user control over visibility, competition, and sharing within gamified mental health systems.

4.4 Limitations

As a scoping review, this work provides a snapshot of literature based on the defined inclusion criteria and search strategy. Limiting results to English-language publications may have excluded relevant work published in other languages [7, 31]. In addition, although the search and screening process followed established scoping review guidance, relevant studies may have been missed if terminology, metadata, or indexing practices fell outside the selected keyword set. This limitation is compounded by the rapid evolution of digital mental health applications and the likelihood that newer studies will continue to emerge.

A second limitation concerns scope and generalizability. The review focuses on gamified digital interventions for stressed college students and may not generalize to other populations or clinical contexts. Many included studies also reported short evaluation periods, limiting insight into the long-term sustainability of reported outcomes [3, 55]. Finally, while this review synthesizes patterns related to exploratory play, reflective engagement, and flexible interaction models, the evidence base remains methodologically diverse and often insufficient for isolating causal relationships between specific mechanics and long-term well-being. Longer-term studies and multi-method evaluations will be important for clarifying how play-oriented design strategies contribute to stress management over time [20, 62].

5 Conclusion

This scoping review examined gamified digital mental health applications designed to support stress management among college students. Across the reviewed corpus, approximately 70% of interventions relied on extrinsic motivators such as points, badges, and streaks. These mechanics were frequently discussed in relation to short-term increases in engagement, yet they were less consistently linked to durable stress-management habits [14, 28, 42]. In contrast, studies that described narrative depth, reflective practices, and self-directed engagement more often framed these features as supporting emotionally meaningful interaction rather than mere task compliance [12, 27, 58]. Drawing on play theory and the concept of a bounded playspace, including the magic circle, this review highlights opportunities to design stress-management tools that support exploration, symbolic meaning-making, and emotionally safe rehearsal rather than primarily rewarding task completion [25, 66]. Because many interventions already incorporate therapeutic techniques such as cognitive behavioral practices, play-oriented frameworks can help clarify how these practices might be embedded in experiences that feel personally meaningful rather than purely compliance-driven [6, 20].

Across the three research questions, the synthesis indicates that extrinsic gamification remains the dominant design approach, while play theory is incorporated more selectively. Ethical and design challenges were recurrent across the literature, particularly around privacy, social visibility, and the tension between supportive accountability and performance pressure [38, 59]. Interventions that incorporated narrative and self-reflective elements were more often discussed as supportive of longer-term engagement and perceived value than those relying exclusively on reward loops [12, 42].

The findings also point to a need for evaluation approaches in human-computer interaction that extend beyond usage metrics. While quantitative measures can describe participation, they often fail to capture perceived meaning, emotional resonance, and reflective engagement. Mixed-method approaches that incorporate self-reported emotional outcomes, qualitative reflection, and longitudinal inquiry may provide a more complete account of how play-based interventions

are experienced over time [11, 62]. In-depth interviews, thematic analysis, and longer-term engagement studies can help clarify how users interpret and internalize play-based interventions, and whether these experiences support coping practices beyond initial novelty.

Overall, this review contributes a structured mapping of how gamified mental health interventions for college stress currently operationalize engagement, where they commonly rely on extrinsic reward structures, and where play-oriented elements remain underutilized. Future work should examine play-based strategies over longer durations and across broader cultural contexts and populations, refining how digital interventions can support meaningful and sustained stress-management practices [3, 13].

A PRISMA search

Table 3. PRISMA breakdown of study selection process.

Step	Count
Initial search results	1,611 studies
After duplicates removed	1,300 studies
Screened based on title & abstract	1,300 studies
Excluded at abstract level (e.g., irrelevant topic, no gamification)	989 studies
Full-text articles assessed for eligibility	311 studies
Excluded after full-text review (e.g., no mental health focus, no empirical data)	245 studies
Final number of included studies	66 studies

B Search Queries

C Studies Evaluated with Game Elements Mapped to Play

Table 4. Database-specific search queries used for identifying studies in the scoping review.

Database	Search Query
ACM Digital Library	(gamification OR play OR game-based) AND (stress OR mental health OR well-being) AND (college OR university OR undergraduate)
PubMed	(gamification OR play OR games) AND (stress OR anxiety OR mental health) AND (college students OR university students)
Scopus	(gamification OR play OR games) AND (stress OR mental health) AND (higher education OR college students)
IEEE Xplore	(gamification OR gameful design) AND (stress OR mental health) AND (students OR higher education)
Taylor & Francis	(gamification OR play) AND (stress OR mental health) AND (college OR university)
EBSCO	(gamification OR play OR games) AND (stress OR well-being) AND (college students OR higher education)
ScienceDirect	(gamification OR play) AND (stress OR mental health) AND (college students OR university students)
SpringerLink	(gamification OR play) AND (stress OR mental health) AND (higher education OR students)

Table 5. Game Elements Reported Across Included Studies Mapped to Whitton's Framework of Play

Game Element	Whitton Type	Play Framing of Engagement (Author-Reported)	How the Element Was Described	Element Was Described in Studies (n=66)	Representative References
Streak Tracking	Structured Play	Primarily extrinsic	Described as encouraging short-term adherence; frequently discussed alongside novelty effects, performance pressure, or disengagement following streak loss	15	Deterding et al. [18]; Mekler et al. [42]; Rodrigues et al. [55]; Looyestyn et al. [38]
Progress Tracking	Structured Play	Extrinsic	Framed as supporting routine participation and monitoring; often discussed in relation to compliance rather than reflection	30	Mohr et al. [46]; Wong et al. [67]
Badges	Structured Play	Extrinsic	Discussed as motivational incentives; several studies reported mixed responses or limited long-term engagement	26	Mekler et al. [42]; Looyestyn et al. [38]
Points	Structured Play	Extrinsic	Described as reinforcing task completion and goal attainment; commonly paired with progress indicators	25	Mekler et al. [42]; Mekler et al. [41]
Daily Check-ins	Exploration Play	Mixed (extrinsic and intrinsic)	Described as supporting low-pressure reflection when framed flexibly; discussed as compliance-oriented when paired with performance metrics	12	Bopp et al. [10]
Quest / Mission Structure (Narrative Tasks)	Exploration Play	Mixed (extrinsic and intrinsic)	Described as task-based engagement structures guiding participation through discrete objectives (e.g., missions, challenges, or quests)	17	Ottam et al. [49]; Rodrigues et al. [55]
Avatar Customization	Imagination Play	Intrinsic	Framed as supporting personal expression and identity-related engagement; typically optional rather than central	7	Reay et al. [53]; Schmidt-Kraepelin et al. [56]
Chatbot Interaction	Exploration Play	Mixed	Described as blending structured guidance with conversational flexibility	6	Fitzpatrick et al. [20]; Mohr et al. [46]
Cooperative Challenges / Social Support	Chal- Social Play	Primarily intrinsic (cooperative)	Discussed in relation to accountability and emotional support; less frequently associated with performance pressure	9	Litvin et al. [36]; Oakley-Girvan et al. [47]
Competitive Features	Social Social Play	Extrinsic (comparative)	Frequently discussed in relation to performance comparison and pressure; several studies raised ethical concerns	15	Landers et al. [30]; Spahl et al. [59]

Note: Counts reflect the number of *unique studies* (out of 66) in which each analytically defined element was identified. Because terms such as *quests*, *missions*, and *challenges* were used inconsistently across studies, these mechanics were merged when they served the same functional role in structuring engagement. Counts therefore reflect normalized analytic categories rather than raw author-reported labels.

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